q7_redesign.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

unsigned long checks = 0;

int is_prime_v1(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    for (unsigned long long i = 2; i * i <= n; i++) { checks++; if (n % i == 0) return 0; }
    return 1;
}

int is_prime_v2(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    if (n == 2) return 1;
    checks++;
    if (n % 2 == 0) return 0;
    for (unsigned long long i = 3; i * i <= n; i += 2) { checks++; if (n % i == 0) return 0; }
    return 1;
}

int main(void)
{
    unsigned long tickets[] = {97, 100, 7919, 1000003, 999983};
    printf("      n   v1 divisions   v2 divisions\n");
    for (int k = 0; k < 5; k++) {
        unsigned long n = tickets[k], a, b;
        int p1 = is_prime_v1(n); a = checks;
        int p2 = is_prime_v2(n); b = checks;
        printf("%8lu  %12lu  %12lu  %s\n", n, a, b, p1 == p2 ? "same result" : "MISMATCH");
    }

    unsigned long t1 = 0, t2 = 0; int bad = 0;
    for (unsigned long n = 1; n <= 100000; n++) {
        int p1 = is_prime_v1(n); t1 += checks;
        int p2 = is_prime_v2(n); t2 += checks;
        if (p1 != p2) bad++;
    }
    printf("\nAll n = 1..100000: v1 = %lu, v2 = %lu divisions, mismatches = %d\n", t1, t2, bad);
    printf("Reduction = %.1f%%\n", 100.0 * (t1 - t2) / t1);
    return 0;
}
```

Terminal - output

```
user@pc:~/Assignment2$ gcc -Wall q7_redesign.c -o q7 && ./q7
      n   v1 divisions   v2 divisions
     97              8              5  same result
    100              1              1  same result
   7919             87             44  same result
1000003            999            500  same result
 999983            998            500  same result

All n = 1..100000: v1 = 2745694, v2 = 1445440 divisions, mismatches = 0
Reduction = 47.4%
user@pc:~/Assignment2$ _
```